

AI SMART CAMPUS MANAGEMENT SYSTEM

Complaint Management System | Digital Notice Board

Major Project Report – BCA612

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Introduction

Traditional campus management relies on paper-based complaints, physical notice boards, and manual workflows.

The AI Smart Campus Management System replaces these inefficient processes with a smart, digital, AI-assisted platform.

The system focuses on two core modules:

1. Complaint Management System with AI Form Generation
2. Digital Notice Board for real-time announcements

It empowers students, staff, and administrators to communicate, resolve issues, and share information seamlessly.

Project Scope

- ✓ Web-based platform
- ✓ 3 User Roles
- ✓ AI-Powered Forms
- ✓ Real-Time Notices
- ✓ MySQL Backend

Institution

Maharishi University of Information Technology

Noida – 201304

BCA 2023-26 Batch

Problem Statement

Challenges in Traditional Campus Management



Manual Complaints

Students visit offices physically. No tracking, no digital record of submitted issues.



Physical Notice Boards

Outdated, damaged or inaccessible. Announcements reach students too slowly.



No Status Updates

Students have no way to check if their complaint is being resolved.



No AI Assistance

Drafting formal complaints is time-consuming for students with no formal writing experience.



No Central Dashboard

Admins lack a single view to track, assign, and resolve all campus complaints.



Poor Communication

No centralized digital channel for broadcasting urgent campus announcements instantly.

Objectives

1 Provide an AI-powered complaint form that auto-generates content from user keywords.

2 Allow students and staff to submit, track and view complaint status in real time.

3 Enable admins to post, edit and manage digital notices campus-wide.

4 Implement Role-Based Access Control (Student / Staff / Admin).

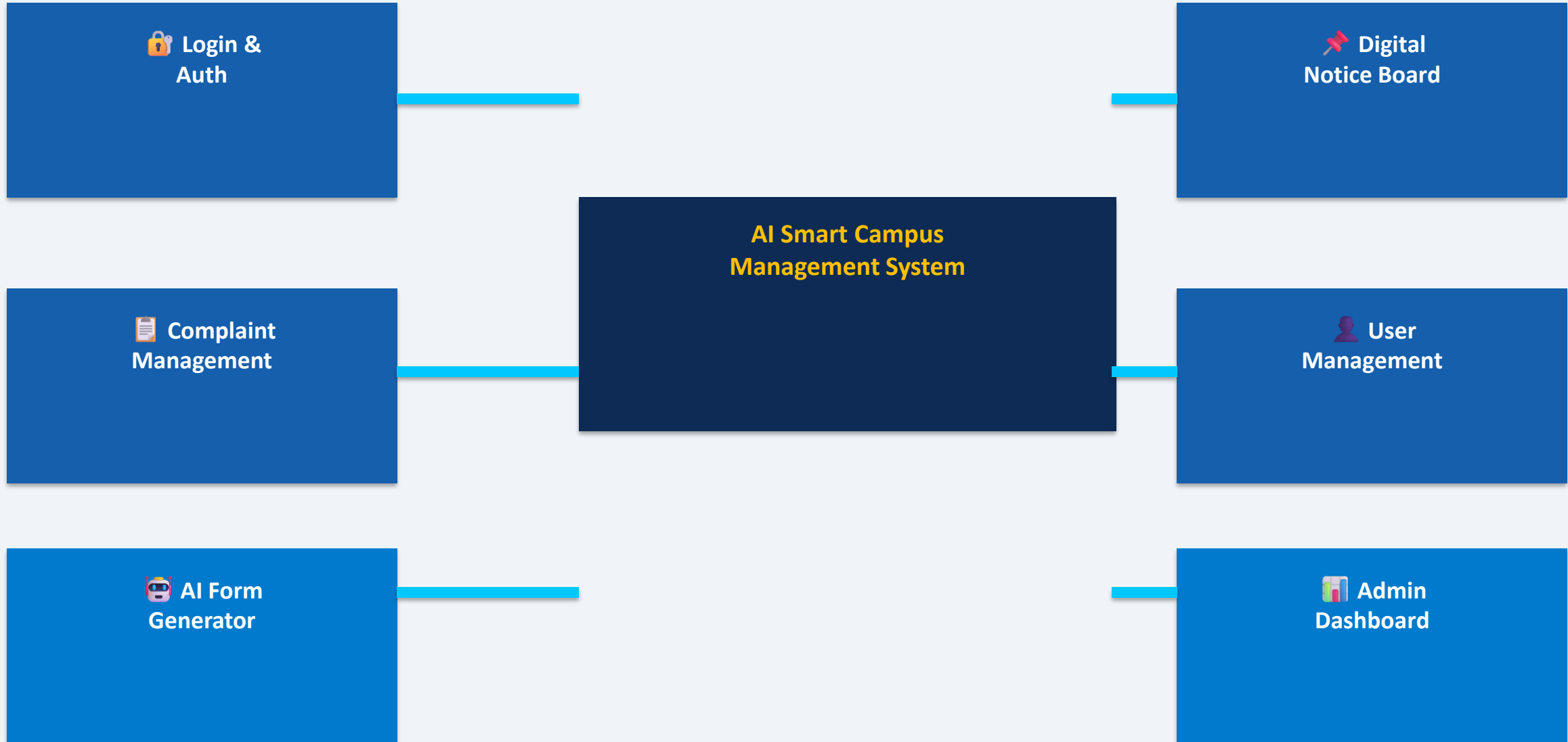
5 Replace physical notice boards with a searchable, categorized digital board.

6 Maintain a complaint workflow: Pending → In Progress → Resolved.

7 Provide search & filter on both complaints and notices.

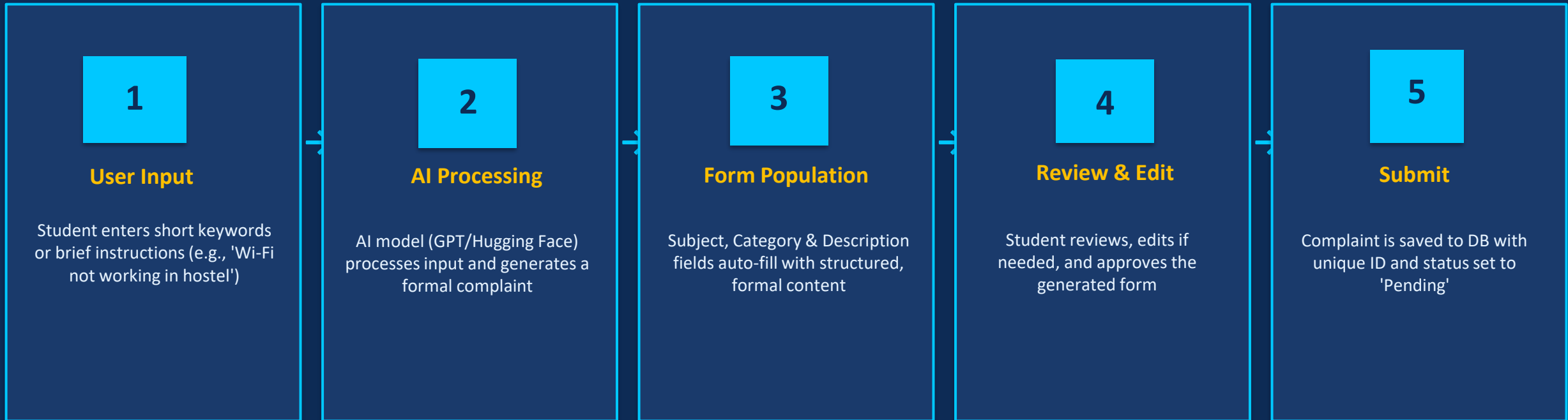
8 Deliver a fully responsive design for desktop, tablet & mobile.

Proposed System Overview



AI Complaint Form Generator

Core Innovation — Auto-fill complaint forms using AI



Instructions Format supported by AI: "[Issue] + [Location] + [Duration] + [Impact]" → e.g. 'Water supply issue in Boys Hostel Block-B for 2 days affecting 80 students'

Digital Notice Board

Post Notices

Admins create notices with title, content, category and expiry date instantly.

Search & Filter

Students filter notices by category (Academic, Event, Hostel, IT, Other) or keyword.

Schedule Notices

Set future publish date – notices go live automatically on the scheduled time.

Priority Levels

Mark notices as Normal, Important or Urgent with color-coded visual badges.

Responsive View

Notice board adapts to mobile, tablet and desktop screens seamlessly.

Auto-Expiry

Expired notices are automatically removed from the board on the set expiry date.

Role-Based Access Control

Two distinct user roles with tailored features

STUDENT

- ✓ Register & Login
- ✓ Submit Complaints (Manual / AI)
- ✓ Track Complaint Status
- ✓ View Digital Notice Board
- ✓ Search & Filter Notices

ADMIN

- ✓ Full Complaints Dashboard
- ✓ Assign Complaints to Staff
- ✓ Post / Edit / Delete Notices
- ✓ Manage All Users & Roles
- ✓ View Analytics & Reports

System Architecture & Data Flow

PRESENTATION LAYER (HTML5 / CSS3 / Bootstrap / JavaScript)



APPLICATION LAYER (Python – Flask Framework / REST API Endpoints)

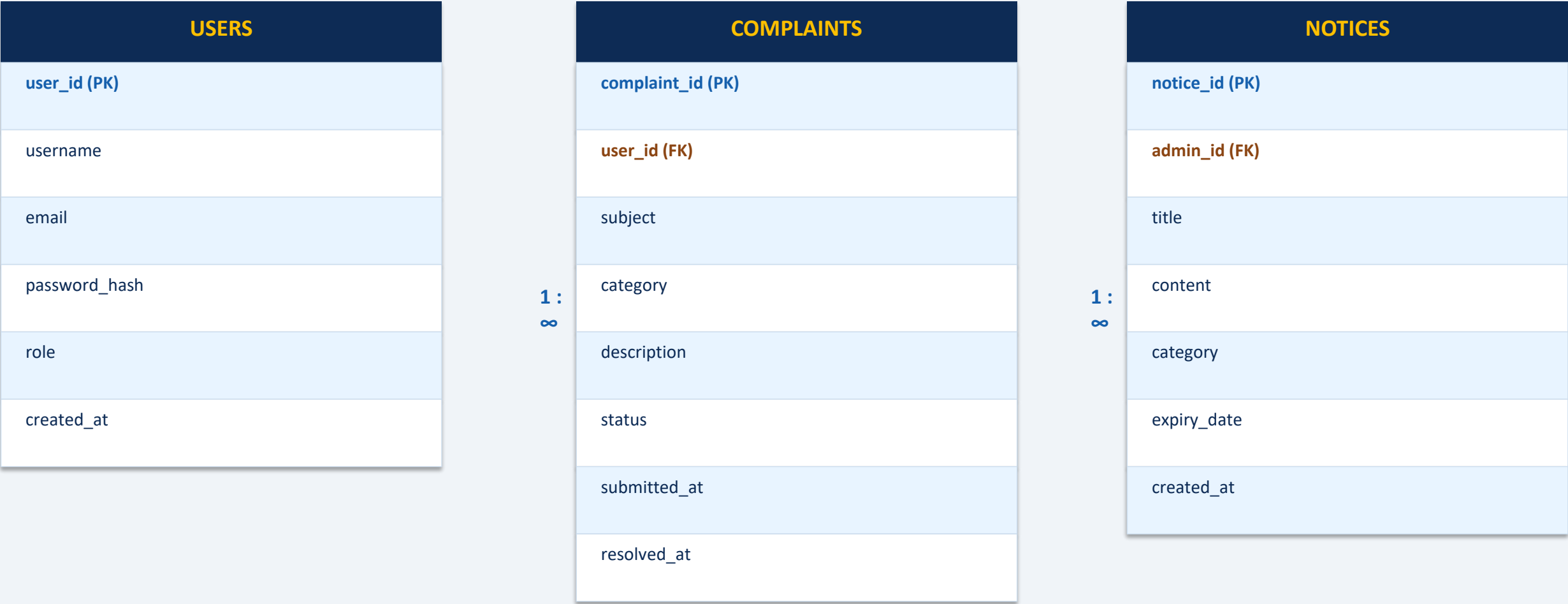


AI MODULE (OpenAI GPT / Hugging Face Transformers – Complaint Form Generator)



DATA LAYER (MySQL Database – Users / Complaints / Notices / AI Logs)

Database Design



USER submits COMPLAINT (1:∞) | USER (Admin) posts NOTICE (1:∞)

Entity-Relationship (ER) Diagram



Complaint Submission Flowchart

Step-by-step process from login to resolution



Technology Stack



Frontend

HTML5, CSS3, JavaScript

Bootstrap 5 / Tailwind CSS

Jinja2 Templates



Backend

Python 3.x

Flask Web Framework

REST API Design



Database

MySQL

SQLAlchemy ORM

Relational Schema Design



AI Module

OpenAI GPT API

Hugging Face Transformers

Instruction-to-Text NLP



Dev Tools

VS Code IDE

Postman (API Testing)

XAMPP Local Server



Deployment

AWS Free Tier / Replit

Apache Web Server

HTTPS (SSL)

Cost & Benefit Analysis

COSTS

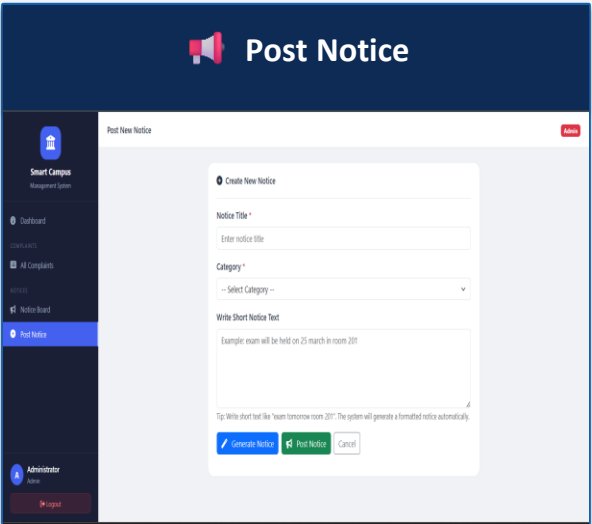
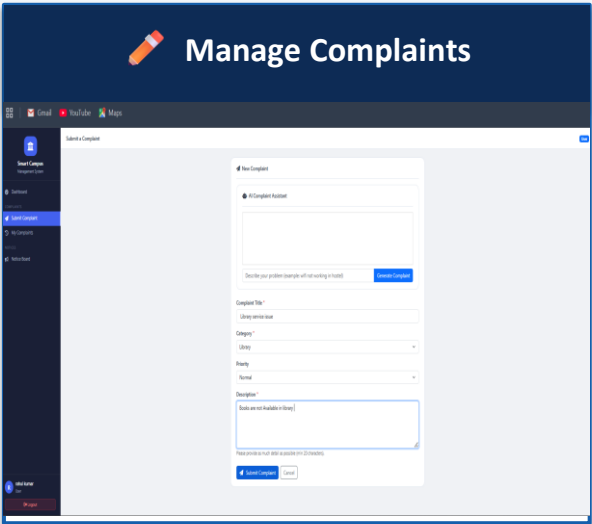
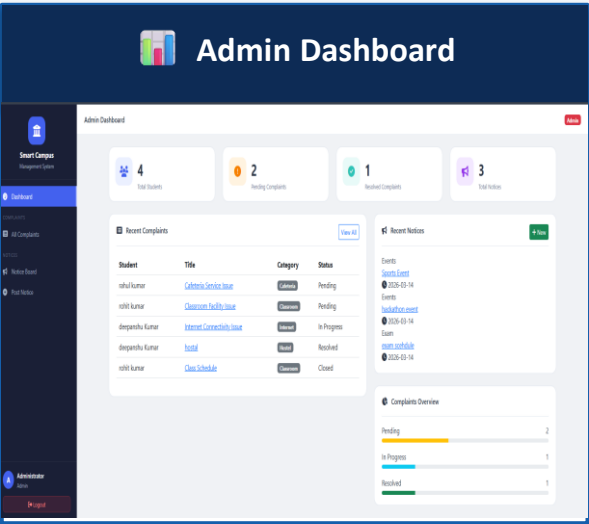
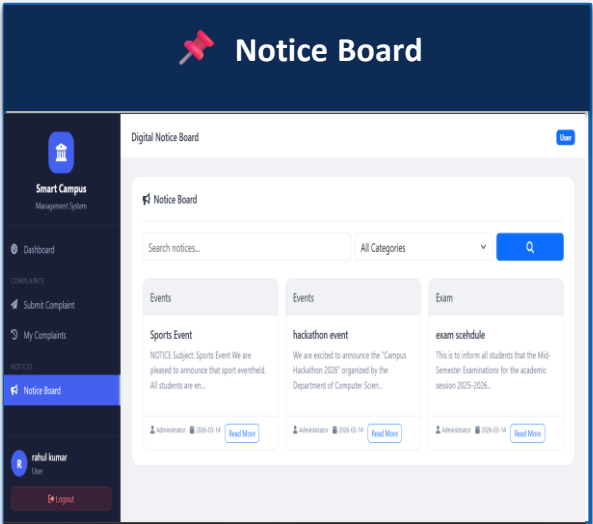
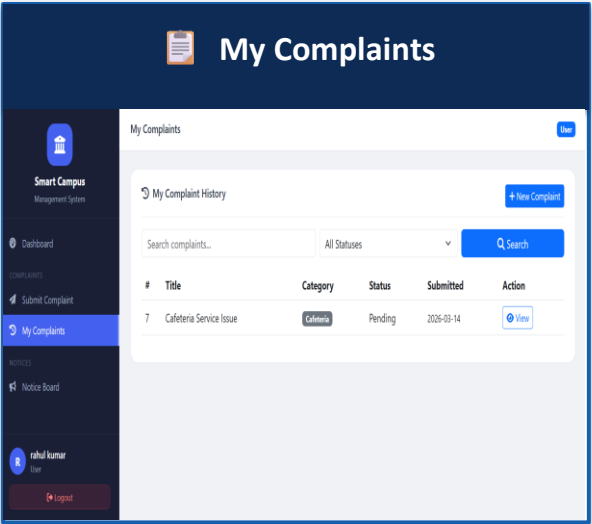
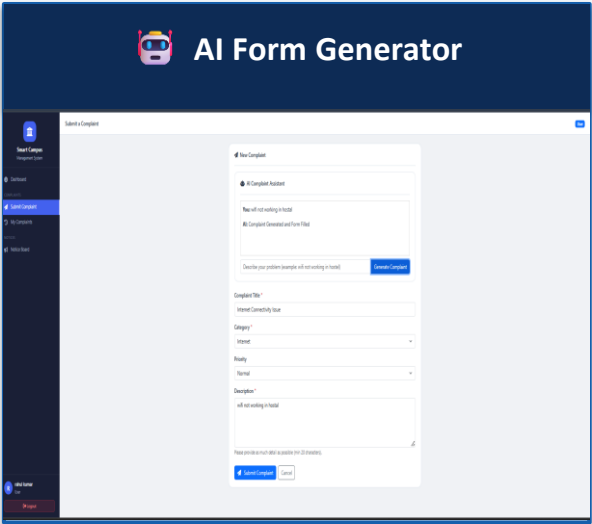
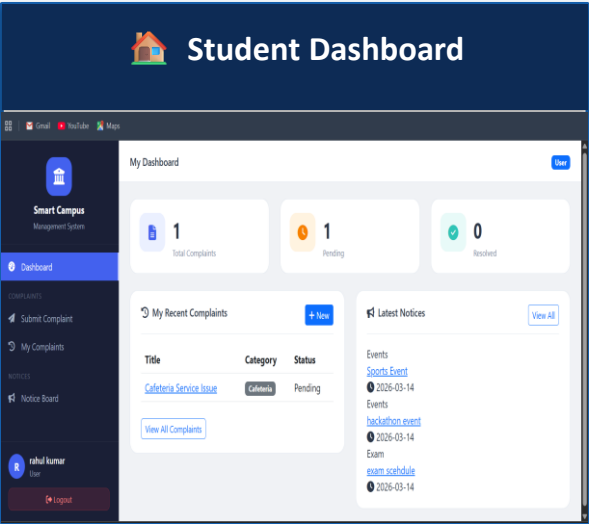
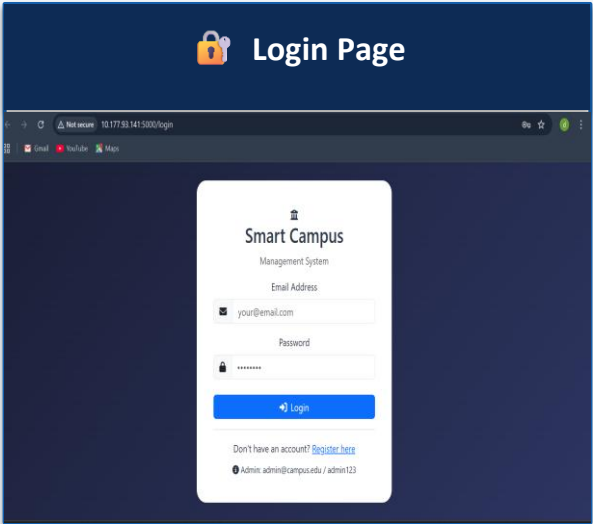
Software Tools	Rs. 0 (Open Source)
AI API	Rs. 0 – Rs. 500 / month
Cloud Hosting	Rs. 0 – Rs. 300 / month
Maintenance	Rs. 500 – Rs. 1000 / month
Development	Low (Academic Project)
Total / Year	~ Rs. 10,000 – 20,000

ROI
POSITIVE

BENEFITS

- ★ 80% reduction in complaint handling time
- ★ 100% paperless complaint & notice process
- ★ Real-time transparency in resolution status
- ★ Instant campus-wide notice dissemination
- ★ Centralized admin control & reporting
- ★ Improved student satisfaction & trust

System Screens / Snapshots



Challenges Faced

Technical, Operational & AI-Specific Challenges



RBAC Implementation

Implementing role-based Flask session validation on every protected route without security gaps.



AI Accuracy

Ensuring AI generates formal, category-relevant complaint text without hallucinating false details.



API Cost Control

Limiting AI API usage to control costs while maintaining quality for all student-submitted forms.



Schema Design

Designing a normalized database handling users, complaints, notices & AI logs without redundancy.



Responsive UI

Making the interface fully responsive across mobile, tablet and desktop views consistently.



Concurrent Users

Handling multiple simultaneous complaint submissions and notice views with optimized DB queries.

Future Scope

Short-Term Enhancements

-  Email / SMS Notifications on status updates
-  Complaint Priority Levels (Low / Medium / High / Critical)
-  Dedicated Android & iOS Mobile Application
-  Scheduled Notice Auto-Publishing
-  Two-Factor Authentication (OTP)
-  Dark Mode / Light Mode Theme Toggle

Long-Term Vision

-  AI Auto-Categorization of Complaints
-  Campus Chatbot for student queries
-  Analytics Dashboard with visual reports
-  Multi-Campus / Multi-Tenant Support
-  Blockchain-based Complaint Integrity
-  Hindi & Regional Language Support

Advantages & Limitations

Advantages

- ✓ AI saves time – complaint forms generated in seconds
- ✓ Fully paperless complaint & notice workflow
- ✓ Real-time complaint status transparency
- ✓ Centralized admin dashboard for full control
- ✓ Responsive – works on all devices
- ✓ Scalable modular architecture
- ✓ Improved campus communication speed
- ✓ Historical data for insights & analytics

Limitations

- ✗ Requires active internet connection
- ✗ AI quality depends on third-party API availability
- ✗ No real-time push notifications yet
- ✗ AI may generate generic text occasionally
- ✗ No dedicated mobile app in current version
- ✗ Single institution – no multi-campus support
- ✗ File upload size limited by server constraints

Conclusion

The AI Smart Campus Management System successfully demonstrates how Artificial Intelligence can transform traditional campus operations into efficient, transparent, and intelligent digital workflows.

By combining an AI-Powered Complaint Form Generator with a Digital Notice Board, the system empowers students, staff, and administrators to work smarter and faster.

This project proves that AI integration in campus management is not just feasible — it is highly beneficial, scalable, and a step toward a truly Smart Campus ecosystem.

THANK YOU

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